

The Milky Way's Hidden Mass: Profiling Gaia DR3

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How the sky is split (HEALPix)

Gaia divides the whole sky into 786,432 tiles using a method called HEALPix (Hierarchical Equal Area isoLatitude Pixelization). Every tile covers the same amount of sky and has a number. Equal areas keep the star counts fair.

How the files are named

Each file holds all the stars in a range of tile numbers, and the file name is that range. Example: GaiaSource_000000-003111 holds tiles 0 to 3111. Files are cut to about the same size (roughly 200 MB), not the same sky area, so a crowded file near the Milky Way needs only a few tiles while an empty file needs thousands. Either way each file holds about 500,000 stars.

Why we took every 28th file

We only need part of the catalogue (over 50 million rows). We chose 121 of the 3,386 files. We took every 28th file, not the first 121, because files are ordered by sky position. The first 121 would all sit in one small patch of sky. Every 28th spreads our sample evenly across the whole sky.

Profiling

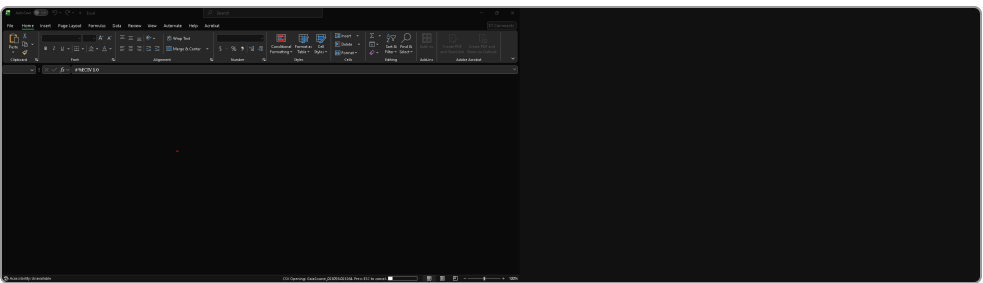
Dataset: ESA Gaia DR3, a full-sky catalogue of about 1.8 billion stars.
Our slice: 121 files, 26 GB compressed (about 120 GB uncompressed), **64,681,399 rows**, 152 columns.

We measured the data with command-line tools that read it line by line, so the file never has to fit in memory. Each command was timed.

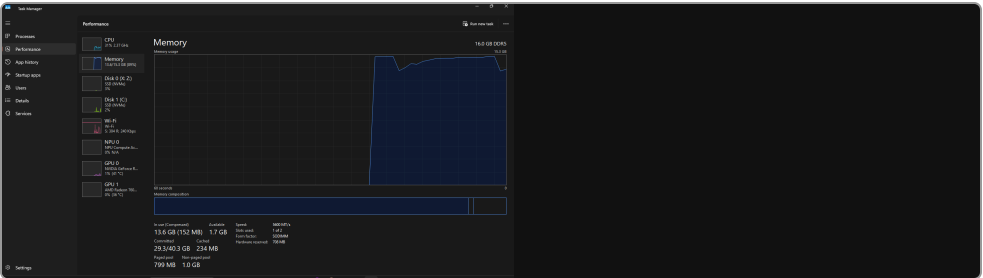
Command (streamed, timed)	Result	Time
du -sh data/	26 GB on disk	0.2 s
zcat *.gz grep -v '^#' wc -l	64,681,399 rows	9 min
... awk average of parallax	0.9428 mas	5 s
... awk sort uniq -c	most stars are faint	5 s
... grep -c VARIABLE	5,345 rows	5 s

Result: memory stayed at a few MB no matter how big the file was, even while reading all 26 GB. The tools stream the data, they never load it. This meets both rules: 64.7 million rows is over 50 million, and 26 GB is over 5 GB.

Breaking



Excel: gets stuck just opening one raw file. Its limit is 1,048,576 rows. Our full data has 64.7 million rows, about **62 times more** than Excel can hold.



pandas: tries to load everything into memory. Loading only one fifth of the data pushed memory use to **29.3 GB**, almost double the 15.3 GB this laptop has (89 percent full, 1.7 GB free). Loading all of it would need about 120 GB, which is **impossible on this laptop**.

Same task: average of parallax	Memory used	Outcome
Command line (zcat awk)	a few MB	0.9428 mas, done
pandas (read_csv)	29.3 GB	ran out of memory

Same laptop, same data, same question. Streaming used a few MB and finished. Loading needed about twice the laptop's memory just to hold one fifth of the rows. That is the breaking point.